Two Pusher Box Sorting System Documentation

🧱 🔴 RUNG 1 — START THE SYSTEM

StartPB AND NOT StopPB AND NOT EstopPB AND NOT Fault → RunLatch

🧠 **Simple Meaning:**

- Press **Start** → machine turns ON
- If **Stop / Emergency / Fault** → machine turns OFF

👉 Think of it like:

"Power switch for the whole factory"

🔴 RUNG 2 — FORCE STOP (RESET RUN)

StopPB OR EstopPB OR Fault → Reset RunLatch

🧠 **Meaning:**

- If anything bad happens → shut everything down

🔴 RUNG 3 — CONVEYOR RUN

RunLatch AND NOT P1_Active AND NOT P2_Active → ConveyorMotor

🧠 **Meaning:**

- Conveyor runs normally
- BUT stops when:
    - Pusher 1 is working
    - OR Pusher 2 is working

👉 Like:

"Don't move the belt while pushing a box"

🔴 RUNG 4 — DETECT NEW BOX (CLASS TRIGGER)

S_Low → R_TRIG

🧠 **Meaning:**

- Detect when a box **enters classification zone**
- Only triggers **once per box**

👉 Like:

"Hey! A new box just arrived!"

🔴 **RUNG 5 — SMALL BOX DETECTION**

R_Class.Q AND S_Low AND NOT S_Mid → Small_Flag

🧠 **Meaning:**

- Only bottom sensor ON → small box

👉 Like:

"This box is short → SMALL"

🔴 **RUNG 6 — MEDIUM BOX DETECTION**

R_Class.Q AND S_Mid AND NOT S_High → Medium_Flag

🧠 **Meaning:**

- Middle sensor ON → medium

👉 Like:

"This one is taller → MEDIUM"

🔴 **RUNG 7 — RESET SMALL FLAG**

P1_Active → Reset Small_Flag

🧠 **Meaning:**

- After using the info → clear it

🔴 **RUNG 8 — RESET MEDIUM FLAG**

P2_Active → Reset Medium_Flag

🔴 **RUNG 9 — DETECT BOX AT PUSHER 1**

S_P1 → R_TRIG

🧠 **Meaning:**

- Detect when box reaches Pusher 1

🔴 **RUNG 10 — COUNT SMALL BOXES**

R_P1.Q AND Small_Flag → Counter++

🧠 **Meaning:**

- When small box reaches P1 → count it

👉 Like:

"We processed one small box"

🔴 **RUNG 11 — ACTIVATE PUSHER 1**

RunLatch AND R_P1.Q AND Small_Flag AND P1_Retracted → P1_Active

🧠 **Meaning:**

- Only push if:
    - System running
    - Box is small
    - Pusher ready

👉 Like:

"Push this box now!"

🔴 **RUNG 12 — PUSH TIMER**

P1_Active → Timer 1s

🧠 **Meaning:**

- Control how long push lasts

🔴 **RUNG 13 — PUSHER OUTPUT**

P1_Active AND NOT TimerDone → P1_On

🧠 **Meaning:**

- While timer running → push forward
- When timer ends → stop → retract

👉 Like:

"Push… then release"

🔴 **RUNG 14 — FINISH PUSH**

TimerDone AND P1_Retracted → Reset P1_Active

🧠 **Meaning:**

- Wait until:
    - Push done
    - Cylinder back

Then:

"Okay, done → resume conveyor"

🔴 **RUNG 15 — DETECT BOX AT PUSHER 2**

Same as P1 but for P2

🔴 **RUNG 16 — COUNT MEDIUM BOX**

R_P2.Q AND Medium_Flag → Counter++

🔴 **RUNG 17 — ACTIVATE PUSHER 2**

Same logic as P1 but for medium boxes

🔴 **RUNG 18 — TIMER P2**

Same idea

🔴 **RUNG 19 — P2 OUTPUT**

Same idea

🔴 **RUNG 20 — COMPLETE P2**

Same idea

🔴 **RUNG 21 — COUNT LARGE BOXES**

R_Class.Q AND NOT Small_Flag AND NOT Medium_Flag → Large_Count++

🧠 **Meaning:**

- If not small or medium → it's large

👉 Large boxes are **not pushed**, only counted

🔴 **RUNG 22 — P1 TIMEOUT**

P1_Active → start 3s timer

🧠 **Meaning:**

- If push takes too long → something is wrong

🔴 **RUNG 23 — P1 FAULT**

TimerDone AND NOT Retracted → Fault

👉 Like:

"Pusher stuck!"

🔴 **RUNG 24–25 — P2 TIMEOUT + FAULT**

Same logic for P2

🔴 RUNG 26 — GLOBAL FAULT

P1_Fault OR P2_Fault → Fault

🔴 RUNG 27–29 — RESET FAULTS

ResetPB → clear all faults

🧠 🔥 COMPLETE STORY (VERY SIMPLE)

1. Press Start → conveyor runs
2. Box enters → classified
3. If SMALL → stop at P1 → push → continue
4. If MEDIUM → stop at P2 → push → continue
5. If LARGE → pass straight
6. Every box counted
7. If something jams → system stops